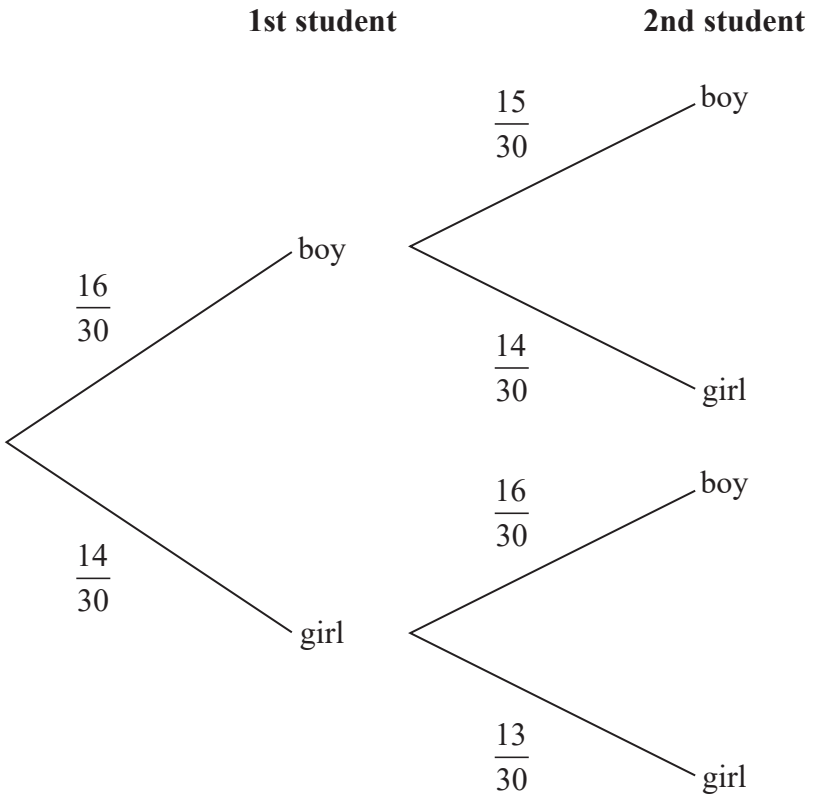
12 There are 30 students in Mr Lear's class.
16 of the students are boys.

Two students from the class are chosen at random.

Mr Lear draws this probability tree diagram for this information.



(a) Write down **one** thing that is wrong with the probabilities in the probability tree diagram.

The second-stage denominators should be 29,

because one student has already been removed.

1 mark
Correct denominator

(1)

Owen and Wasim play for the school football team.

The probability that Owen will score a goal in the next match is 0.4
The probability that Wasim will score a goal in the next match is 0.25

Mr Slater says,

"The probability that both boys will score a goal in the next match is 0.4 + 0.25"

(b) Is Mr Slater right?
Give a reason for your answer.

No. For both, multiply probabilities (assuming independence):

0.4 × 0.25 = 0.10.

1 mark
Both requires multiplication, not addition

(1)

(Total for Question 12 is 2 marks)